Md Mahadi Hasan

3+ years of professional experience; Eager to make bigger impact
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EXPERIENCE

Frontend (React.js) Developer, SegWitz Sdn. Bhd. — Malaysia (Remote) Mar 2023 – Oct 2024

- Developed **data-driven** web interfaces and dashboards supporting business workflows.
- Automated build, testing, and deployment (**CI/CD**) pipelines using **GitHub Actions**, improving development efficiency.
- Optimized document generation and data handling processes, **reducing processing time by 2x**.

Frontend Developer, Tech Joint Bridge — Dhaka, Bangladesh May 2022 – Jan 2023

- Built interactive applications for data-intensive use cases using **React.js** and **Next.js**.
- Implemented data **visualization** and image **processing** pipelines using **Python** and **JavaScript**.
- Improved application performance by **5x**, enabling faster access to analytical data.

Junior Software Developer, Teresa Startup — Dhaka, Bangladesh July 2021 – Jan 2022

- Developed the "**Smart Nurse System**" an innovative digital healthcare system.
- Project secured **government funding**, demonstrating real-world business value.

SKILLS & TECHNOLOGIES

Programming Languages: Python, JavaScript, C++, Java
Frameworks & Libraries: TensorFlow, PyTorch , Keras, SK-Learn, React.js, Next.js
Automation & AI: Machine Learning, Deep Learning, LangChain, RAG, Selenium
Data Analysis & Visualization: Pandas, Numpy, Matplotlib, Seaborn
Web & Mobile Development: HTML, CSS, JavaScript, Semantic Web, Android Studio
Cloud & DevOps: Git, CI/CD, Azure
APIs: REST API, GraphQL
Databases: MySQL, PL/SQL, MongoDB

PROJECTS

AI-Based Embryo Classification for Marine Snails Jan 2025 – Present

- Applied **deep learning** techniques (U-Net, Mask R-CNN, GANs) to classify Littorina saxatilis embryos by developmental stage.
- Designed **hybrid framework** combining segmentation, transfer learning, and generative models for microscopy image analysis.

Healify – AI-Powered Physiotherapy Assistant [Link] Sept 2025 – Dec 2025

- Engineered a comprehensive healthcare platform integrating **LLM** with **RAG** for intelligent medical assistance and real-time computer vision (**MediaPipe**) for automated exercise tracking and posture analysis.
- Led the **design, architecture, and development** of a scalable full-stack system in a **cross-functional** project, using FastAPI, React, and WebSockets for **real-time streaming**, with MongoDB, Azure, and GitHub Actions for **CI/CD** automation.

Multi-Agent Delivery System [Link] May 2025 – May 2025

- Designed and implemented python based **multi-agent** delivery simulation system with rerouting and coordination capabilities.
- Utilized Python **SPADE** framework with agent-based **BDI** architecture, JSON messaging, and **FIPA ACL** communication standards**.**

United Nations (ESG START) [Link] Mar 2023 – Aug 2024

- Developed for the **UN** in Malaysia and used by companies, including **Petronas**. Utilizes **RestAPI, React-chartjs2, CI/CD, Agile**.
- Supports **Localization**, offering navigation in both **English and Malay**, **dynamic PDF** generation using **React.pdf**.

RESEARCH

1. Sentiment Analysis from Bengali Social Media Posts Using Hybridized Feature Extraction Approach**.** [DOI Link]
2. Bengali Social Media Post Sentiment Analysis Using Deep Learning and BERT Model**.** [DOI Link]

EDUCATION

MSc in Artificial Intelligence – University of Jyväskylä Aug 2024 – June 2026

BSc in Computer Science & Engineering – Ahsanullah University of Science and Technology (cgpa: 3.39/4) Mar 2017 – Jan 2022

ACHIEVEMENTS & TRAINING

1. **Devpost Postman Hackathon —** Among 1875 participants worldwide, secured 72nd place for developing Fake News Detector.
2. **B-JET (Bangladesh Japan ICT Engineers' Training Program) – 2022 —** Acquired foundational skills in the Japanese language, along with knowledge of various technologies and business etiquette.